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Technical Communication

BAS301

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** ALL | **Semester:** 3 | **Credits:** 3 | **L-T-P:** 3-0-0

Scheme: AKTU B.Tech 2nd year, 2023-24 onwards (also offered as BAS401 in semester 4)

UNIT 1 Fundamentals of Communication and Voice Dynamics

UNIT 2 Communication Skills for Career Building

UNIT 3 Writing, Designing and Speaking for Presentation

UNIT 4 Communication and Leadership Development

UNIT 5 Digital Communication and Personality

Original revision notes written for the AKTU syllabus. Every unit has:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: lists, models and diagrams that make up long answers.
3. **Solved examples / formats**: ready-to-adapt formats (email, proposal, SOP, CV, pitch) that AKTU asks you to draft.
4. **Key points to remember**: compact lists.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (2 hours):** Unit 1 (communication process diagram, barriers, 7 Cs, flow of communication) and Unit 2 (CV, interview, GD, SOP).
- **Afternoon (2 hours):** Unit 3 (report, proposal, research paper structure, pitching) and Unit 4 (leadership, feedback, emotional intelligence, thinking).
- **Evening (1 to 2 hours):** Unit 5 (emails, blogs, podcasts, personality theories), then practise writing one email and one proposal from the formats.
- **In the exam:** use headings and bullet points, draw the communication cycle and Maslow's pyramid, and follow the exact format when asked to draft something. Invent sensible details (names, dates) when the question says so.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit.

UNIT 1 Fundamentals of Communication and Voice Dynamics

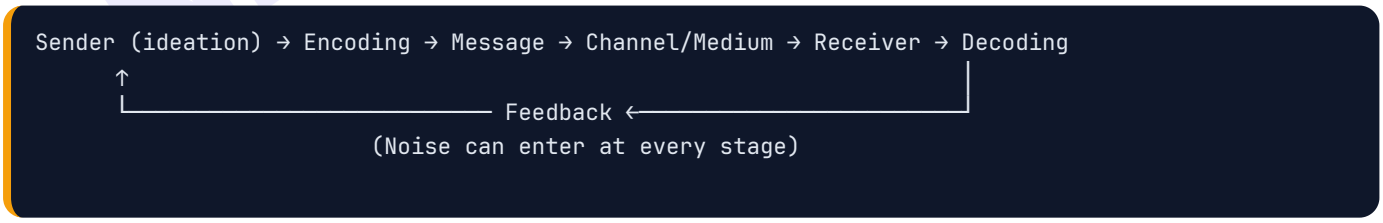
✍️ 2-mark definitions

- **Communication:** the process of sharing information, ideas and feelings between a sender and a receiver through a medium, so that they reach a common understanding. From Latin *communis*, "common".
- **Technical communication:** communicating specialised (scientific, technical, professional) information clearly, accurately and concisely to a specific audience for a specific purpose.
- **Code:** the system of signs used to send a message (language, symbols, gestures). **Content:** the actual message or meaning conveyed.
- **Stimulus and response:** a stimulus is what triggers communication (a need, event or message); the response is the receiver's reaction, which becomes feedback.
- **Feedback:** the receiver's response that tells the sender whether the message was understood; it completes the communication cycle.
- **Noise:** anything that disturbs or distorts the message at any stage.
- **Vowel sounds:** speech sounds made with an open vocal tract (English has 20 vowel sounds: 12 pure vowels and 8 diphthongs). **Consonant sounds:** made by blocking or narrowing the air flow (24 in English).
- **Intonation (tone):** the rise and fall of pitch. **Falling tone** for statements, commands and wh-questions; **rising tone** for yes/no questions, doubt and politeness.

💡 Core concepts

Purposes / roles of communication: to inform, to instruct, to persuade, to motivate, to build relationships, to make decisions, to express feelings, to coordinate work.

Communication process (draw as a cycle)



Types of communication

Basis	Types
Medium	Verbal (oral, written) and non-verbal (body language, gestures, paralanguage, proxemics, chronemics)
Formality	Formal (official channels) and informal (grapevine)
Direction / flow	Downward, upward, horizontal (lateral), diagonal (cross-wise)

Basis	Types
Audience	Intrapersonal, interpersonal, group, mass

Flow of communication in an organization

Flow	From → to	Examples
Downward	Superiors → subordinates	Orders, policies, instructions
Upward	Subordinates → superiors	Reports, suggestions, grievances
Horizontal	Same level	Coordination between departments
Diagonal	Different levels and departments	Engineer talking directly to accounts manager
Grapevine	Informal, any direction	Rumours, informal news

Formal communication networks: chain, wheel (centralized, one leader in the middle), circle, Y-network, all-channel (star, everyone talks to everyone).

Barriers to effective communication

Type	Examples
Physical / environmental	Noise, distance, poor network, bad lighting
Semantic / language	Jargon, ambiguous words, poor translation
Psychological	Emotions, prejudice, poor attention, stress
Socio-cultural	Different customs, values, gestures
Organizational	Rigid hierarchy, too many levels, information overload
Personal	Poor listening, lack of confidence, attitude

Overcoming barriers: simple and clear language, know the audience, active listening, use feedback, choose the right channel, empathy, avoid information overload.

7 Cs of effective communication: Clarity, **Conciseness**, **Concreteness**, **Correctness**, **Coherence**, **Completeness**, **Courtesy** (some books use *Consideration* instead of Coherence).

🧮 Solved examples

Example 1 (barrier analysis). A project lead emails the client a status update full of internal acronyms; the client misunderstands the delay. Barrier: **semantic** (jargon). Fix: explain terms, use the 7 Cs (clarity, completeness), and invite questions.

Example 2 (intonation).

- "We have finished the report." ↘ (falling: statement)
- "Have you finished the report?" ↗ (rising: yes/no question)
- "Where is the report?" ↘ (falling: wh-question)

Key points to remember

```
Process: sender → encode → message → channel → receiver → decode → feedback (noise anywhere)
Flows: downward, upward, horizontal, diagonal, grapevine
Networks: chain, wheel, circle, Y, all-channel
Barriers: physical, semantic, psychological, socio-cultural, organizational
7 Cs: clear, concise, concrete, correct, coherent, complete, courteous
Falling tone: statements, wh-questions;  rising tone: yes/no questions
```

⭐ Important questions

1. Define communication and explain its process with a diagram (cyclic nature). *Label every element including noise and feedback.*
2. Explain the types and the flow of communication in an organization.
3. Classify the barriers to effective communication and suggest ways to overcome them.
4. Explain the 7 Cs of communication with examples.
5. Write notes on code and content, stimulus and response, and intonation (rising and falling tone).

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UNIT 2 Communication Skills for Career Building

✍️ 2-mark definitions

- **CV (curriculum vitae):** a detailed record of one's academic and professional history; longer, used for academic/research roles.
- **Résumé:** a short (1–2 page) summary of skills, education and experience tailored to a particular job.
- **Interview:** a formal, purposeful conversation in which an employer assesses a candidate's suitability.
- **Group discussion (GD):** a structured discussion among a group on a topic, used in selection to judge communication, leadership, teamwork and reasoning.
- **Effective profiling:** presenting oneself well across platforms (résumé, LinkedIn, portfolio, GitHub) with a consistent professional image.
- **Networking:** building and maintaining professional relationships that create opportunities and support.
- **Statement of Purpose (SOP):** an essay for higher-studies applications explaining one's background, goals and why one fits the programme.
- **Expert technical lecture:** a talk by a specialist explaining a technical subject to an audience, which needs clarity and structure.

💡 Core concepts

CV vs résumé

CV	Résumé
Detailed, 2+ pages	Brief, 1–2 pages
Full academic and professional history	Selected, relevant information
Same for every application	Tailored for each job
Academic, research, international roles	Industry jobs

Résumé sections: contact details → career objective/summary → education → skills (technical and soft) → projects/internships → achievements/certifications → extracurriculars → declaration (optional). Use action verbs ("designed", "led"), quantify results, keep it error-free.

Interview skills

- **Before:** research the company, revise your résumé, prepare answers (tell me about yourself, strengths/weaknesses, why us), dress formally.
- **During:** arrive early, firm greeting, good **body language** (eye contact, upright posture, open hands, smile), listen fully, answer with examples (STAR: Situation, Task, Action, Result), be honest.
- **After:** thank the panel, send a follow-up email.
- **Types:** personal, telephonic/video, panel, technical, HR, stress, group interview.

Why GDs are used in recruitment: they test, in a short time and in a group setting, communication, listening, leadership, teamwork, logical thinking, knowledge and confidence under pressure. **GD dos:** start or summarize if you can, give facts and examples, let others speak, disagree politely, stay on topic. **Don'ts:** shout, interrupt repeatedly, dominate, stay silent throughout.

SOP elements: introduction with a hook → academic background → relevant projects/research/internships → why this field → why this university/programme → career goals → conclusion. Be specific, personal and honest; usually 800–1000 words.

Seminar and conference paper structure: Title → Authors and affiliation → Abstract → Keywords → Introduction → Literature review → Methodology → Results and discussion → Conclusion and future scope → References (IEEE/APA style).

Expert technical lecture (writing and presenting): know the audience, clear objective, logical outline (introduction, body, conclusion), simple visuals, real examples, clarity of terms, time management, Q&A at the end. **Clarity matters** because the audience may not share the expert's background; unclear jargon loses them.

🧮 Solved examples (formats)

Résumé summary line: "Final-year B.Tech (CSE) student with hands-on experience in Python and React, built a library management system used by 300 students; seeking a software developer role."

Answer to "Tell me about yourself" (structure): present (who you are now) → past (education, key project) → future (why this role).

SOP opening: "The day our college website crashed during admissions, I spent a night learning how load balancing works. That curiosity has shaped my journey towards a master's in distributed systems..."

Key points to remember

```
CV = detailed, academic;  résumé = short, tailored
Interview: prepare → body language → STAR answers → follow-up
GD tests: communication, leadership, teamwork, reasoning
SOP: hook, background, projects, why field, why university, goals
Paper: title, abstract, keywords, intro, literature, method, results, conclusion, references
```

⭐ Important questions

1. Differentiate CV and résumé; draft a résumé for a given job.
2. Explain interview skills, including the role of body language.
3. What is a group discussion? Why is it used in recruitment? Give dos and don'ts.
4. Explain the elements of a Statement of Purpose; write an SOP outline.
5. Explain the structure of a research paper for a seminar/conference.
6. Explain how to write and present an expert technical lecture; why is clarity important?
7. Explain effective profiling and networking for career building.

UNIT 3 Writing, Designing and Speaking for Presentation

✍️ 2-mark definitions

- **Report:** a factual, structured document presenting information, analysis and findings on a topic, usually for decision-making.
- **Proposal:** a persuasive document that suggests a plan or project and asks for approval or funding.
- **Thesis:** a long research document submitted for a degree, presenting original research.
- **Project report:** a document describing a completed project: objectives, method, results and conclusions.
- **Pitch:** a short, persuasive presentation of an idea to investors, managers or clients.
- **Elevator pitch:** a 30–60 second summary of who you are or what your idea is, short enough to deliver in a lift ride.
- **Elements of speech delivery: passion** (genuine enthusiasm), **poise** (calm confidence and control), **illustrations** (examples, stories, visuals).

💡 Core concepts

Report vs proposal

Report	Proposal
Describes what was done or found	Suggests what should be done
Informative, factual	Persuasive
Based on past/present	About the future
Ends with findings and recommendations	Ends with request for approval/funding

Structure of a project report / thesis

1. Title page, certificate, declaration, acknowledgement
2. Abstract
3. Table of contents, list of figures/tables
4. Introduction (background, problem statement, objectives, scope)
5. Literature review
6. Methodology / system design
7. Implementation
8. Results and discussion
9. Conclusion and future scope
10. References, appendices

Structure of a technical proposal: title → introduction/background → problem statement/need → objectives → proposed plan/methodology → schedule (timeline) → resources and budget → expected

outcomes/benefits → evaluation → conclusion and request.

How to pitch an idea (process, preparation, structure)

1. Know your audience and their interests.
2. Hook: a striking fact, story or question.
3. Problem: what pain point exists.
4. Solution: your idea and how it works.
5. Market / benefit: who gains and how much.
6. Proof: prototype, data, testimonials.
7. Team and plan: who will do it, timeline, money needed.
8. Call to action: the clear ask, then thank and take questions.

Elevator pitch vs business pitch

Elevator pitch	Business pitch
30–60 seconds	10–20 minutes, often with slides
Spark interest, get a follow-up meeting	Get investment or approval
Who you are, what you do, why it matters, ask	Problem, solution, market, model, finance, team, ask

Designing presentation slides: one idea per slide, 6×6 rule (max ~6 lines, ~6 words each), large readable fonts, consistent theme, visuals over text, cite sources.

🧮 Solved examples (formats)

Example 1 (proposal: tree plantation drive).

- **To:** The Principal, ABC Institute of Technology. **From:** Eco Club. **Date:** 5 July 2026. **Subject:** Proposal for a campus and neighbourhood tree-plantation drive.
- **Background:** campus temperature and dust levels have increased; green cover is below 15%.
- **Objective:** plant 500 native saplings on campus and in the nearby colony by 15 August 2026.
- **Plan:** survey sites (week 1), collect saplings from the forest department nursery (week 2), plantation day with 200 student volunteers (week 3), assign each sapling to a student "guardian" for one year.
- **Budget:** saplings ₹10,000 (subsidised), tree guards ₹25,000, tools and water tanker ₹8,000; total ₹43,000.
- **Benefits:** cooler campus, cleaner air, student involvement, NAAC/community-service credit.
- **Request:** approval and funds of ₹43,000 from the institute's green fund.

Example 2 (elevator pitch). "Hi, I'm Riya, a CSE student. Every semester, students waste hours searching for verified notes. We built *NoteNest*, an app that organises AKTU notes by subject and unit, already used by 2,000 students. I'd love 15 minutes to show you how it could work for your coaching centre."

Key points to remember

```
Report = facts, past/present;  proposal = persuasion, future
Project report: abstract → intro → literature → method → results → conclusion → references
Proposal: background, need, objectives, plan, schedule, budget, benefits, request
Pitch: hook → problem → solution → benefit → proof → plan → ask
Delivery: passion, poise, illustrations
```

⭐ Important questions

1. Explain the structure of a project report / thesis.
2. Write a technical proposal for a given situation (invent details). *Follow the format in Example 1.*
3. Differentiate a report and a proposal.
4. Explain how to pitch an idea; compare an elevator pitch and a business pitch; outline the steps.
5. Explain the elements of effective speech delivery: passion, poise and illustrations.

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UNIT 4 Communication and Leadership Development

✍️ 2-mark definitions

- **Leadership communication:** the way a leader shares vision, inspires, guides and listens to a team to achieve goals.
- **Social competence:** the ability to interact effectively with others, combining cognitive, emotional and behavioural skills in the right context.
- **Feedback:** information about performance given to help improve it. **Constructive feedback** is specific, timely, balanced and focused on behaviour, not personality.
- **Tact:** the skill of saying the right thing in the right way at the right time without offending. **Intelligence:** the ability to learn, reason and solve problems. Tact is social skill; intelligence is mental ability.
- **Emotional intelligence (EI):** the ability to recognise, understand and manage one's own emotions and those of others (Daniel Goleman).
- **Thinking skills:** mental processes we use to make sense of information, solve problems and make decisions.

💡 Core concepts

Traits of a good leader: vision, integrity, clear communication, empathy, decisiveness, accountability, confidence, adaptability, ability to delegate, active listening, inspiring others.

Leadership communication: share a clear vision; be transparent and honest; listen actively; give recognition and feedback; adapt style to audience; lead by example; communicate in crises calmly.

Communication and social competence: four components

Component	Meaning
Context	Understanding the situation, setting and audience
Feelings	Awareness of one's own and others' emotions
Intentions	The purpose behind what we say
Behaviours	What we actually say and do

Skills needed: cognitive (perspective taking, problem-solving), emotional (empathy, emotion regulation), behavioural (verbal and non-verbal skills, cooperation, conflict resolution). Good communication builds trust, friendship and teamwork, so it improves social competence.

Giving and receiving feedback

- **Giving:** be specific, timely, private for criticism, use "I" statements, balance positives and improvements (sandwich method), suggest solutions.
- **Receiving:** listen without interrupting, don't be defensive, ask for clarification, thank the person, act on it.

Emotional intelligence: four components (Goleman's model, simplified)

1. **Self-awareness:** knowing your emotions and their effect.
2. **Self-management (self-regulation):** controlling impulses and staying calm.
3. **Social awareness (empathy):** understanding others' feelings.
4. **Relationship management (social skills):** influencing, teamwork, resolving conflict.

(Goleman's five-element version adds **motivation**.) EI builds **trust through communication** because people feel understood and safe.

Types of thinking

Type	Meaning	Everyday example
Critical	Analysing and evaluating evidence	Checking if a news story is fake
Creative	Generating new ideas	Designing a poster for a fest
Analytical	Breaking a problem into parts	Debugging code step by step
Logical	Reasoning from premises to conclusion	Solving a puzzle
Lateral	Solving problems indirectly/creatively (de Bono)	Finding a new route during a road block
Convergent / divergent	One best answer / many possible answers	MCQ vs brainstorming

Tools of thinking: mind maps, brainstorming, SWOT analysis, six thinking hats, flowcharts, questioning (5 Whys).

🧮 Solved examples

Example 1 (constructive feedback). Instead of "Your presentation was bad", say: "Your data slides were strong. The introduction ran for five minutes, which lost the audience; try cutting it to one minute next time."

Example 2 (tact vs intelligence). A team member proposes a weak idea. An intelligent person sees the flaw; a tactful person says, "Interesting thought; how might we handle the cost issue?", pointing out the flaw without hurting them.

Key points to remember

```
Leader traits: vision, integrity, communication, empathy, decisiveness, accountability
Social competence: context, feelings, intentions, behaviours
Feedback: specific, timely, balanced (sandwich), behaviour-focused
EI: self-awareness, self-management, social awareness, relationship management (+ motivation)
Thinking: critical, creative, analytical, logical, lateral, convergent/divergent
```

⭐ Important questions

1. What is leadership communication? List the traits of a good leader.

2. Explain communication and social competence: context, feelings, intentions and behaviours; list the skills needed.
3. Explain how to give and receive feedback effectively.
4. Differentiate tact and intelligence with examples.
5. What is emotional intelligence? Explain its components and how it builds trust.
6. Explain thinking and its tools; illustrate five types of thinking used in everyday life.

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UNIT 5 Digital Communication and Personality

✍️ 2-mark definitions

- **Digital communication:** exchanging information through digital platforms: email, social media, blogs, podcasts, webinars, video calls.
- **Email:** an electronic message sent over the internet; the main medium of formal digital communication.
- **Blog:** a regularly updated website or page of articles written in an informal or conversational style.
- **Podcast:** a series of audio (or video) episodes that listeners can download or stream on demand.
- **Webinar:** a live, interactive online seminar where the audience can ask questions. **Webcast:** a one-way live broadcast over the internet with little or no interaction.
- **Personality:** the unique, relatively stable pattern of thoughts, feelings and behaviour that makes a person distinct.
- **Introvert / extrovert (Jung):** introverts draw energy from their inner world; extroverts from the outer world and people.

💡 Core concepts

Webinar vs webcast vs podcast

	Webinar	Webcast	Podcast
Format	Live video seminar	Live broadcast	Recorded audio/video episodes
Interaction	Two-way (Q&A, polls)	One-way	One-way, on demand
Audience	Small–medium	Large	Subscribers, anytime

Writing effective emails: clear subject line, proper salutation, purpose in the first line, short paragraphs, polite tone, call to action, closing and signature; check spelling and attachments; avoid all caps and slang; reply promptly.

Content creation for social media: know the audience, define the goal, use visuals and short text, consistent posting, engage with comments, add value (tips, how-tos), use hashtags wisely, measure results.

Effective and ethical use of social media: verify before sharing (avoid fake news), respect privacy and copyright, credit sources, avoid hate speech and cyberbullying, keep a professional tone on professional platforms, protect personal data, disclose sponsorships.

Speech and personality: voice, pitch, pace, clarity and vocabulary shape how others see our personality; confident, clear speech projects credibility.

Personality theories

- **Maslow's hierarchy of needs (pyramid, bottom to top):** 1. Physiological (food, sleep) → 2. Safety (security, job) → 3. Love/belonging (friends, family) → 4. Esteem (respect, recognition) → 5. **Self-**

actualization (realising full potential). Lower needs must be met before higher ones motivate us; personality grows as we move up.

- **Freud:** personality has three parts: **id** (instincts, pleasure principle), **ego** (reality principle, mediator), **superego** (morals, conscience). Conflicts between them are handled by defence mechanisms.
- **Swami Vivekananda (Pancha Kosha, five sheaths of the self):** 1. **Annamaya** (physical body), 2. **Pranamaya** (vital energy/breath), 3. **Manomaya** (mind, emotions), 4. **Vijnanamaya** (intellect, wisdom), 5. **Anandamaya** (bliss, the innermost self). Personality development means developing all five, from the physical to the spiritual.
- **Carl Jung's typology:** two attitudes, **introversion** and **extraversion**, and four functions: **thinking, feeling, sensation, intuition**, giving eight psychological types. This is the basis of the **MBTI** (16 types) used in personality assessment.

Types of personality (common classifications): introvert/extrovert/ambivert; Type A (competitive, time-conscious) and Type B (relaxed, patient); Big Five (OCEAN: openness, conscientiousness, extraversion, agreeableness, neuroticism).

🧮 Solved examples (formats)

Example 1 (email: invitation to a blood donation camp).

```
To: all-staff@abcit.edu.in; students@abcit.edu.in
Subject: Invitation: Blood Donation Camp on 20 March 2026, Main Auditorium

Dear Faculty Members and Students,

I am pleased to invite you to the Blood Donation Camp organised by our NSS unit
in association with the District Hospital Blood Bank.

Date: 20 March 2026   Time: 10:00 am to 3:00 pm   Venue: Main Auditorium

Donors must be 18–65 years old, weigh at least 50 kg and should have had a meal
before donating. Please register at the NSS desk or through the link below by 18 March.

One unit of blood can save up to three lives. Let us make this camp a success.

Warm regards,
Dr. A. K. Sharma
Director, ABC Institute of Technology
```

Example 2 (blog post outline): catchy title → hook → 3–4 subheaded sections with tips → images → conclusion with a question to readers → tags.

Key points to remember

```
Webinar (live, two-way) vs webcast (live, one-way) vs podcast (recorded, on demand)
Email: subject, salutation, purpose first, short, polite, call to action, signature
Maslow: physiological → safety → belonging → esteem → self-actualization
Freud: id, ego, superego
Vivekananda: annamaya, pranamaya, manomaya, vijnanamaya, anandamaya
Jung: introvert/extrovert × thinking, feeling, sensation, intuition → MBTI
```

⭐ Important questions

1. Explain strategies for writing effective emails; draft an email for a given situation. *Use the format in Example 1.*
2. Distinguish webinars, webcasts and podcasts; explain content creation for blogs and social media.
3. Explain the effective and ethical use of social media.
4. Explain Maslow's hierarchy of needs in relation to personality. *Draw the pyramid.*
5. Explain Swami Vivekananda's five-self (Pancha Kosha) theory of personality development.
6. Explain Freud's and Jung's theories of personality and Jung's typology for personality assessment.

Last-hour checklist

Read only this in the last hour before the paper.

- ☐ Communication cycle diagram with noise and feedback.
- ☐ Flows: downward, upward, horizontal, diagonal, grapevine; networks.
- ☐ Barriers (5 types) and remedies; 7 Cs.
- ☐ Rising vs falling tone.
- ☐ CV vs résumé; interview body language; GD purpose; SOP elements.
- ☐ Research paper and project report structures.
- ☐ Report vs proposal; proposal format.
- ☐ Pitch steps; elevator vs business pitch.
- ☐ Leader traits; social competence; feedback; tact vs intelligence.
- ☐ EI four components; types of thinking.
- ☐ Email format; webinar vs webcast vs podcast.
- ☐ Maslow, Freud, Vivekananda (5 koshas), Jung.

Books prescribed by AKTU

These are the books named in the AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	Technical Communication: Principles and Practice	Meenakshi Raman, Sangeeta Sharma	Oxford University Press	WorldCat
2	Personality Development and Soft Skills	Barun K. Mitra	Oxford University Press	WorldCat
3	Technical Communication (6th ed.)	William S. Pfeiffer	Pearson	WorldCat

#	Title	Author(s)	Publisher	Where to get it legally
4	Soft Skills and Employability Skills	Sabina Pillai, Agna Fernandez	Cambridge University Press	WorldCat
5	Practical Communication: Process and Practice	L. U. B. Pandey	A.I.T.B.S. Publishers	WorldCat

Official syllabus: AKTU B.Tech 2nd year CSE syllabus, 2023-24 scheme (PDF)

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